

Splunk Mastery

From Data Ingestion to SOC Insights

Learning Summary

I learned Splunk architecture, data input types, and the full data pipeline from ingestion to search. I practiced key SPL commands (search, where, stats, eval, etc.) and used the Job Inspector for optimization. I worked with knowledge objects (macros, tags, event types, calculated fields, field extractions) and managed their settings. I created dashboards, reports, alerts, and workflow actions, and built data models with acceleration for better performance by covering end-to-end Splunk functionality for real SOC use. I believe that with more practice, my ability to work effectively in a SOC environment will continue to grow.

Splunk Environment Overview

- Worked in a simulated Splunk environment for hands-on training.
- Utilized the Splunk Cloud Platform (14-day free trial) for lab exercises.
- **Mac Log** files were sourced from GitHub for analysis.
- **Index** in Splunk was labeled "main".
- **Source** was set to Mac logs.
- **Sourcetype** defined as syslog.
- Environment designed to replicate real-world enterprise logging scenarios.

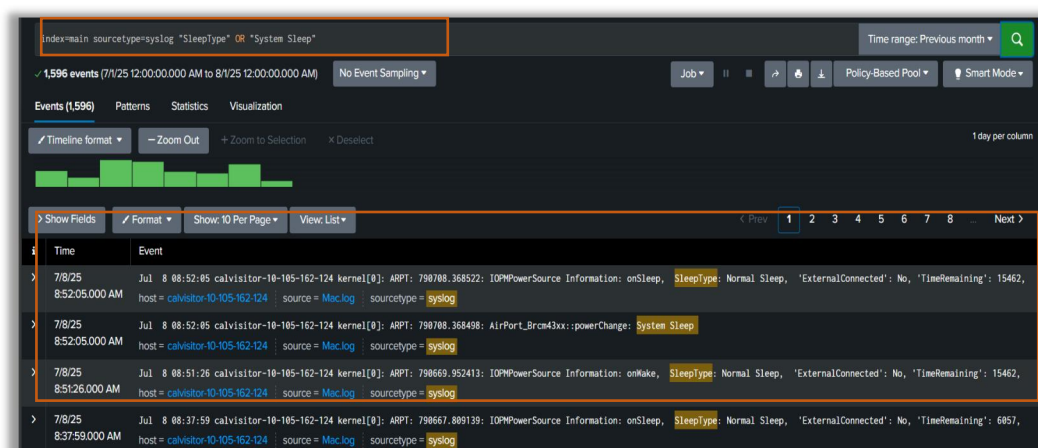
Practical Part

1. Search Query

Used SPL to filter, analyze, and retrieve relevant data from indexed events.

Events

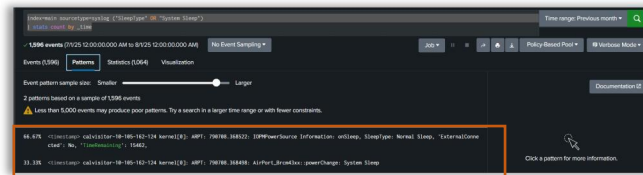
I executed a query and reviewed the raw events it returned to understand the exact data records.



Findings: I examined raw syslog entries containing "sleep type" to verify relevant sleep state logs with timestamps.

Patterns

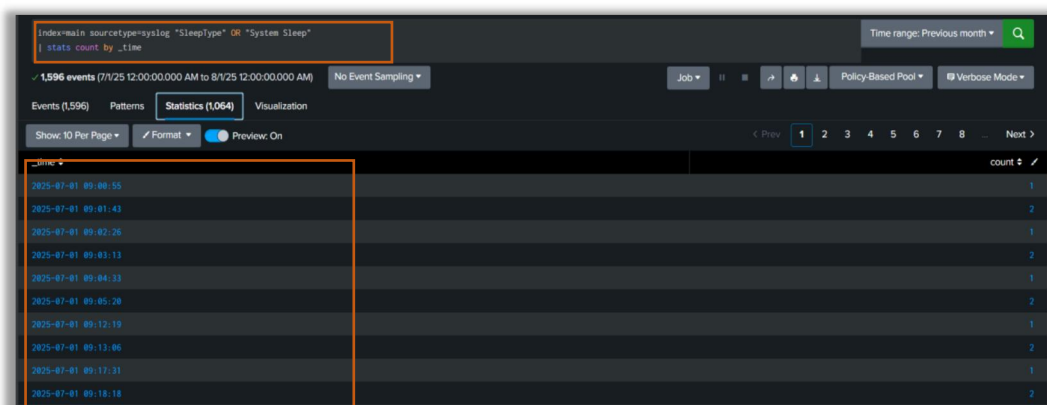
I analyzed the recurring patterns within those events to quickly identify similarities and anomalies.



Findings: I aggregated sleep events over time using stats count by `_time` to measure event frequency during the observed period.

Statistics

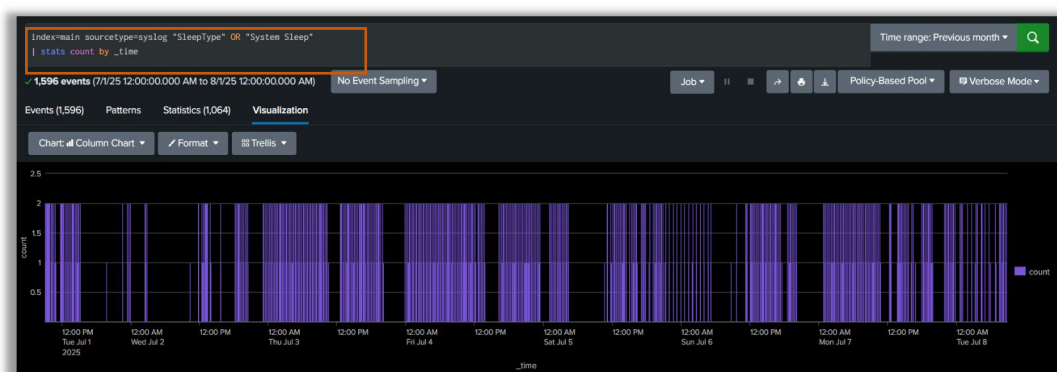
I generated statistical summaries from the query results to observe trends, counts, and relationships in the data.



Findings: The time-based counts showed regular sleep cycles with some anomalies and irregularities in sleep behavior.

Visualization

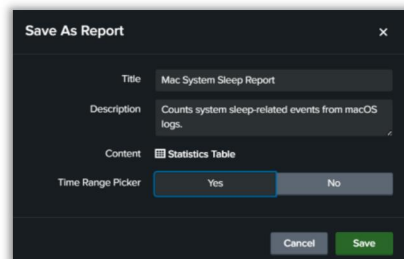
Represented search results through charts for better insights.



Findings: A time chart was used to clearly display sleep event trends and periods of system-inactivity.

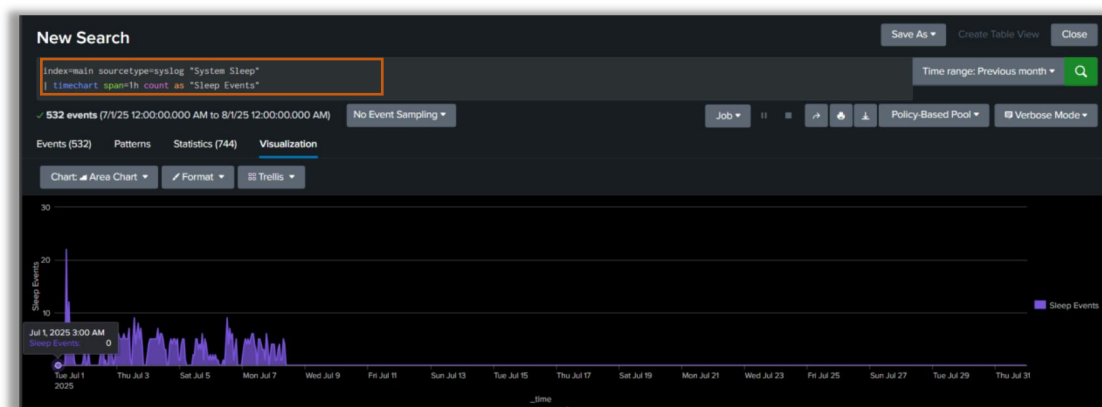
2. Report

I created reports to save and share search results, enabling scheduled execution and consistent access to important data insights.



3. Dashboard

- Designed interactive panels for real-time monitoring and visual analytics.
- This dashboard shows macOS sleep events (like “System Sleep” and “SleepType”) over time using an area chart. The wave patterns represent when and how often sleep events occur.
- It helps monitor power behavior, spot unusual activity, and analyze user/system sleep trends.



Save Panel to New Dashboard

Dashboard Title: System Sleep Activity
system_sleep_activity Edit ID

Description: Optional

Permissions: Private

Dashboard type: Classic Dashboards (The traditional Splunk dashboard builder) or Dashboard Studio (A new builder to create visually-rich, customizable dashboards)

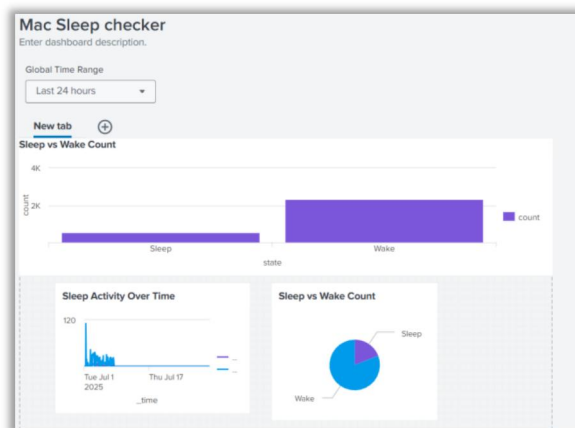
Panel Title: Optional

Visualization Type: Statistics Table

Cancel Save to Dashboard

Dashboard Studio: This dashboard provides a real-time overview of system sleep and wake events on macOS logs. It includes three visualizations:

- **Sleep vs Wake Events (Pie Chart):**
Displays the proportion of sleep vs wake events to identify user/device behavior trends.
- **Sleep Activity Over Time (Line Chart):**
Shows when and how frequently the system went into sleep mode, helping identify usage patterns or possible issues.



4. Alert

I configured alerts to automatically notify when specific search conditions are met, helping in proactive monitoring and quick incident response.

Alerts can be send in different ways:

Alerts

List in interface
Log events
Output to lookup
Send to a telemetry endpoint
Trigger scripts
Send emails
Use a webhook

The 'Save As Alert' dialog box contains the following fields and options:

- Title:** High Sleep Events Alert
- Description:** Triggers when system sleep events exceed threshold.
- Permissions:** Private (selected), Shared in App
- Alert type:** Scheduled
- Run every day:** (selected)
- At:** 2:00
- Expires:** 24 hours
- Trigger Conditions:**
 - Trigger alert when:** Number of Results
 - is greater than:** 5
 - Trigger:** Once (selected), For each result
- Buttons:** Cancel, Save

The alert configuration page shows the following details:

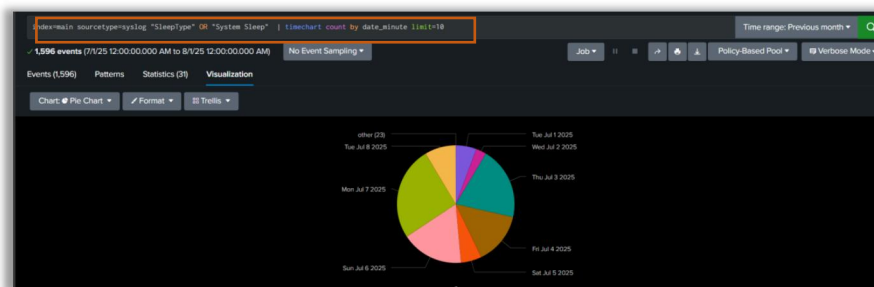
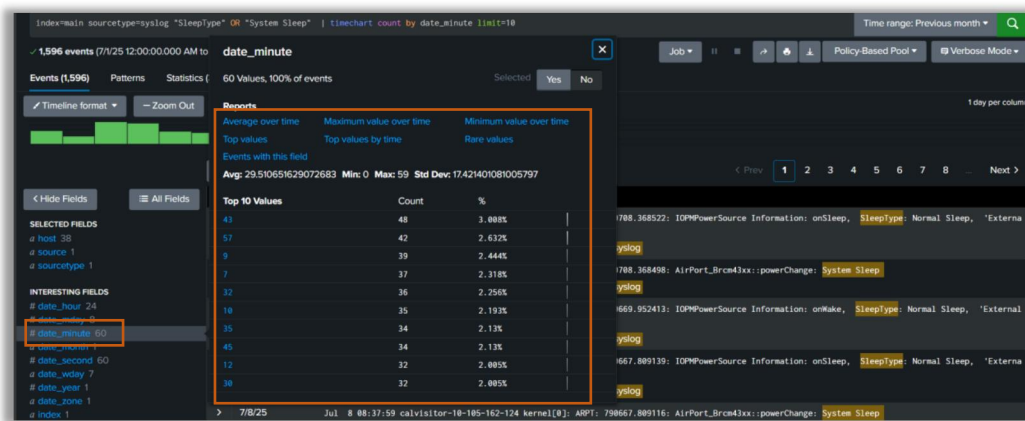
- Alert type:** Scheduled
- Run every day:** (selected)
- At:** 2:00
- Expires:** 24 hours
- Trigger Conditions:**
 - Trigger alert when:** Number of Results
 - is greater than:** 5
 - Trigger:** Once (selected), For each result
- Throttle:** ☐
- Trigger Actions:**
 - + Add Actions:** (button)
 - When triggered:** Send email (selected)
- Buttons:** Cancel, Save

The alert details page for 'High Sleep Events Alert' shows the following information:

- Title:** High Sleep Events Alert
- Description:** Triggers when system sleep events exceed threshold.
- Enabled:** Yes, [Disable](#)
- App:** search
- Permissions:** Private, Owned by sc_admin, [Edit](#)
- Modified:** Aug 8, 2025 4:44:06 AM
- Alert Type:** Scheduled, Daily, at 2:00, [Edit](#)
- Trigger Condition:** Number of Results is > 5, [Edit](#)
- Actions:** 1 Action, [Edit](#)
- Action:** Send email

5. Fields

Specific attributes extracted from events that store related data values. I explored individual fields to understand their distinct values, patterns, and relevance for analysis.



Findings: The query retrieves the latest 10 sleep-related events to provide a quick snapshot of recent system power state changes. This helps in monitoring the most current sleep activity for timely analysis and troubleshooting.

6. Fieldsummary

A quick overview of all extracted fields in the current dataset, showing field names, values, and their frequency for deeper insights.

field	count	distinct_count	is_exact	max	min	numeric_count	stdev	values
date_hour	1596	24	1	23	11.86281887518787	0	1596	6.312333895961886
date_mday	1596	8	1	8	4.3888972431877694	1	1596	2.941683744749885
date_minute	1596	60	1	59	29.510651629072683	0	1596	17.421401081005797

Findings: This provided insights into the frequency and types of sleep events, highlighting key patterns in the system's power management behavior.

7. Knowledge Objects

Saved reusable search elements like fields, event types, and tags for efficiency.

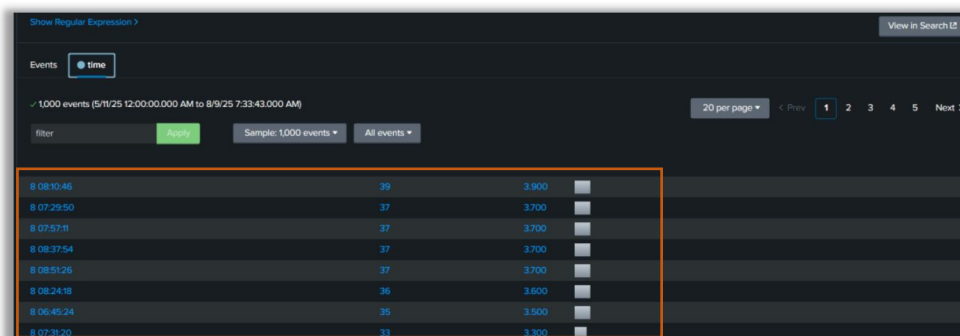
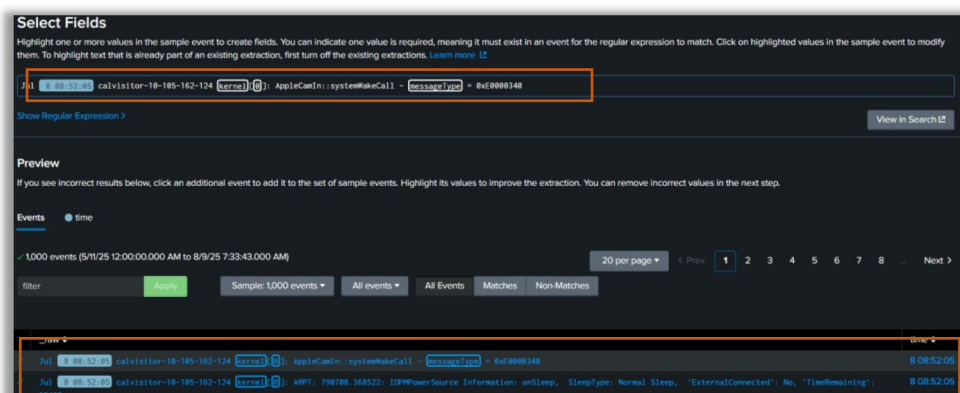
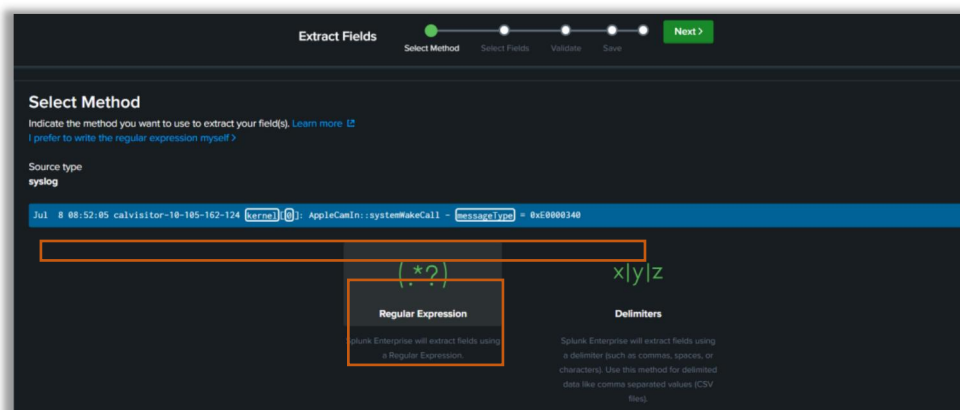
Different Knowledge Objects are following:

Field Extraction

Process of identifying and extracting fields from event. I performed it to make unstructured data searchable.

Two ways for Field Extraction:

- **Regular Expression Field Extraction:** Used regex patterns to precisely identify and extract specific data fields from raw events.



Save
Name the extraction and set permissions.

Extractions Name:

Owner:

App:

Permissions:

Source type:

Sample event:

Fields:

Regular Expression:

- **Delimiter-Based Field Extraction:** Extracted fields by splitting event data using defined delimiters such as commas, spaces, or tabs.

Select Method
Indicate the method you want to use to extract your field(s). [Learn more](#) [I prefer to write the regular expression myself](#)

Source type:

Regular Expression

Splunk Enterprise will extract fields using a Regular Expression.

Delimiters

Splunk Enterprise will extract fields using a delimiter (such as commas, spaces, or characters). Use this method for delimited data like comma separated values (CSV files).

Rename Fields
Select a delimiter. In the table that appears, rename fields by clicking on field names or values. [Learn more](#)

Delimiter:

field1	field2	field3	field4	field5
Jul 8 08:24:58 calvisitor-10-105-162-124 kernel[0]: ARP: 790660.790843: ICMPPowerSource Information: onSleep	onSleep	Normal Sleep	ExternalConnected: No	TimeRemaining: 15484

Preview (5 fields)

Events: ☒ field1 ☐ field2 ☐ field3 ☐ field4 ☐ field5

✓ 1,000 events (5/11/25 12:00:00.000 AM to 8/9/25 7:42:52.000 AM)

filter Sample: 1,000 events

Jul 8 02:41:24 calvisitor-10-105-163-10 kernel[0]: ARP: 720413.310484: ICMPPowerSource Information: onSleep	1	0.203
Jul 8 02:54:51 calvisitor-10-105-163-10 kernel[0]: ARP: 720415.503655: ICMPPowerSource Information: onWake	1	0.203
Jul 8 02:54:51 calvisitor-10-105-163-10 kernel[0]: RTC: Maintenance 2017/7/5 09:54:50	1	0.203

Save
Name the extraction and set permissions.

Extractions Name:

Owner:

App:

Permissions:

Source type:

Sample event:

Fields:

Delimiter:

Field Alias

Created alternate names for existing fields to make searches more intuitive and improve query flexibility.

Add new

Fields
Field aliases
Add new

Destination app
search

Name *
system_power

Apply to
sourcetype
named *
syslog

Field aliases
sleeptype
=
sleep
Delete

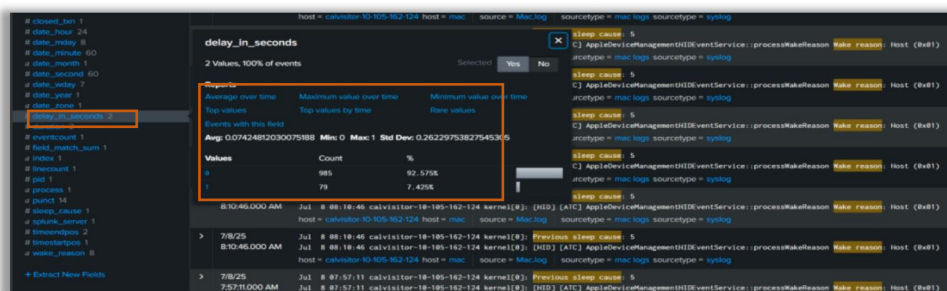
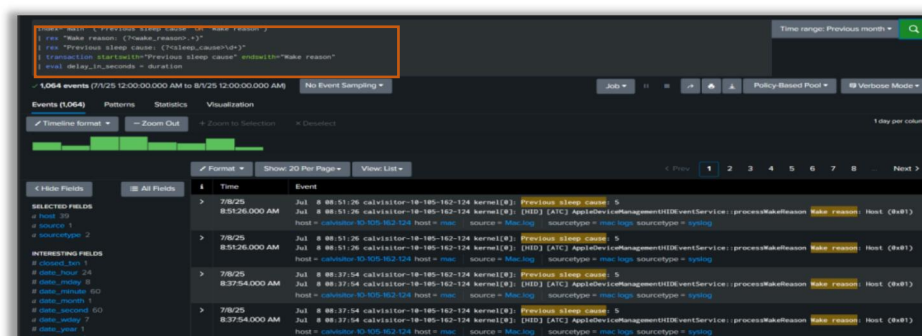
+ Add another field

☐ Overwrite field values

Cancel
Save

Calculated Field

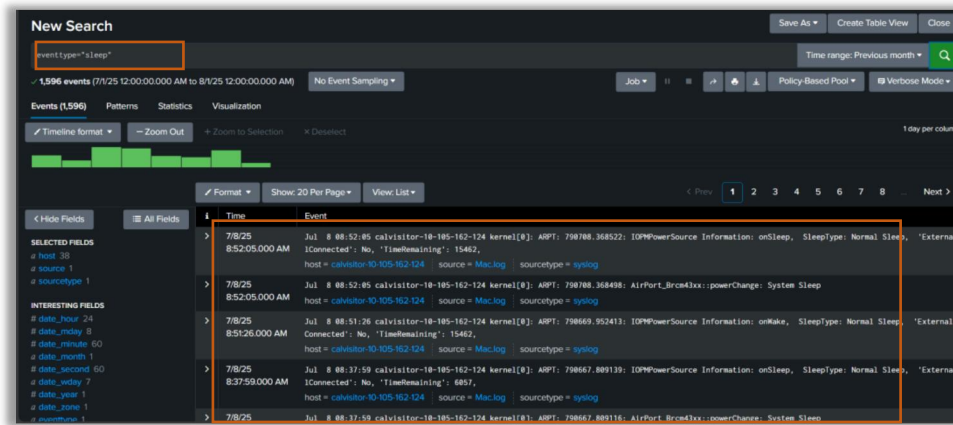
Created new derived fields using expressions and functions in SPL without changing raw data.



Findings: The query extracted detailed logs from the “main” index showing previous sleep causes, wake reasons, and sleep durations. This helps diagnose power management issues by pinpointing exact triggers and duration of system sleep cycles.

Event Type

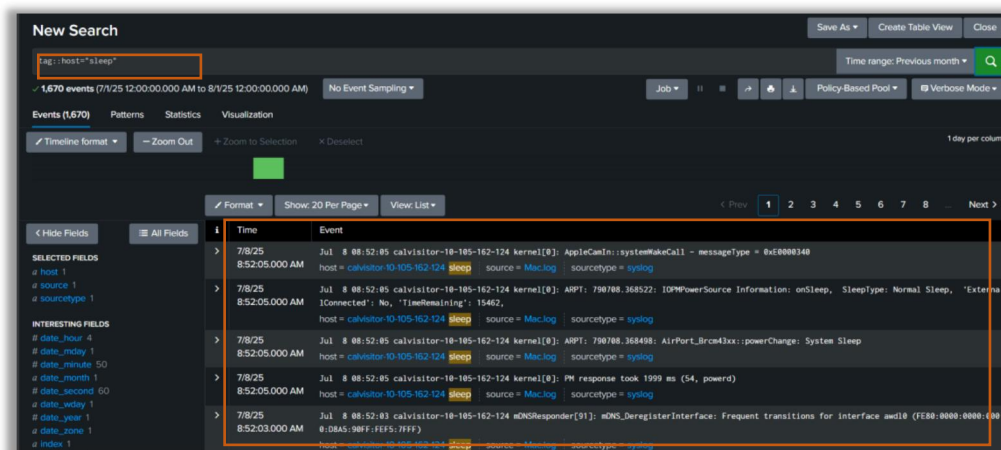
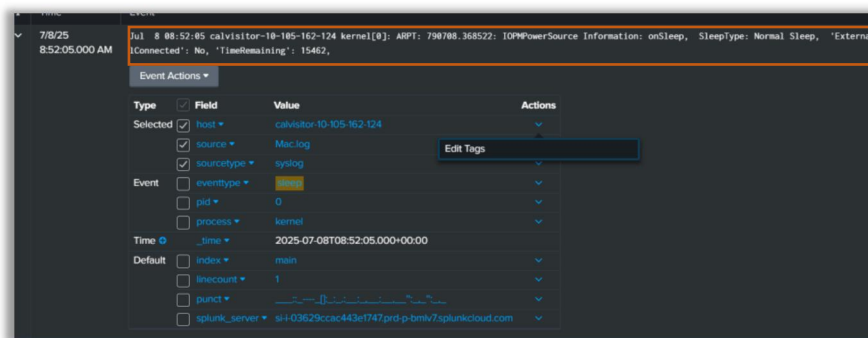
Grouped events with common characteristics into reusable definitions.



Findings: The event type effectively categorizes sleep-related logs, enabling efficient analysis of system power state transitions.

Tag

Assigned labels to related events or fields to enhance search and categorization.



Findings: A sleep-related tag was created to easily identify and group all logs associated with system sleep and wake events for streamlined analysis.

Macro

Built reusable search code blocks to simplify complex queries.

The screenshot shows a 'Macro' configuration window. It includes fields for 'Destination app' (set to 'search'), 'Name' (set to 'power_state[0]'), 'Definition' (set to 'index="main" host="calvisitor-10-105-162-124" | stats count BY power_state'), 'Arguments' (empty), 'Validation Expression' (set to 'state="sleep" OR state="wake"'), and 'Validation Error Message' (set to 'Please provide a valid state value: "sleep" or "wake" only.'). There are 'Cancel' and 'Save' buttons at the bottom right.

Findings: The macro efficiently summarizes powerstate counts, aiding quick analysis of system sleep and wake patterns.

8. Data Models

Designed structured datasets for faster and more organized data analysis.

- **Datasets:** These are essentially **collections of data** that you can search, transform, and visualize. They're like organized views of your indexed data.
- They can be of different types, such as:

Event Dataset: Created a root dataset representing a core event search for pivoting.

The screenshot shows the 'Add Event Dataset' interface. It includes a 'Data Model' dropdown set to 'Mac Sleep Logs'. There are input fields for 'Dataset Name' (set to 'Event dataset for logs') and 'Dataset ID' (set to 'Event_dataset_for_logs'). A 'Constraints' field contains the query 'index="main" sourcetype=syslog host="calvisitor-10-105-162-124"'. Below these fields, there is a 'Sample' dropdown set to '1000 events'. At the bottom, there is a table of events with columns for time, host, and message. The table shows three rows of log data.

Built specialized datasets inheriting from a root event dataset for targeted analysis.

The screenshot shows the 'Add Child Dataset' configuration window. The 'Dataset Name' is 'child dataset of logs' and the 'Dataset ID' is 'child_dataset__of_logs'. The 'Inherit From' dropdown is set to 'Event dataset for logs'. The 'Additional Constraints' field contains 'onSleep'. The 'Event' table at the bottom shows two rows of data with the 'onSleep' field highlighted in orange.

Event
Jul 8 08:52:05 calvisitor-10-105-162-124 kernel[0]: ARP: 790708.368522: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': No, 'TimeRemaining': 15462,
Jul 8 08:37:59 calvisitor-10-105-162-124 kernel[0]: ARP: 790667.809139: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': No, 'TimeRemaining': 6057,

Transaction Dataset: Built a dataset that groups related events into single transactions based on shared fields and time constraints, enabling end-to-end activity tracking.

The screenshot shows the 'Add Transaction Dataset' configuration window. The 'Dataset Name' is 'Sleep Wake Sessions' and the 'Dataset ID' is 'Sleep_Wake_Sessions'. The 'Group Datasets' field contains '-child dataset of logs'. The 'Group by' field contains 'host'. The 'Duration' fields are 'Max Pause: Hours' and 'Max Span: Hours'. The 'Event' table at the bottom shows four rows of data with the 'onSleep' field highlighted in orange.

Event
Jul 8 05:35:34 calvisitor-10-105-162-124 kernel[0]: ARP: 785169.443126: IOPMPowerSource Information: onSleep, SleepType: Standby, 'ExternalConnected': No, 'TimeRemaining': 599,
Jul 8 06:37:41 calvisitor-10-105-162-124 kernel[0]: ARP: 787923.262510: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': Yes, 'TimeRemaining': 57,
Jul 8 07:27:36 calvisitor-10-105-162-124 kernel[0]: ARP: 790457.689442: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': No, 'TimeRemaining': 362,
Jul 8 07:30:49 calvisitor-10-105-162-124 kernel[0]: ARP: 790519.297042: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': No, 'TimeRemaining': 10909,

Findings: The datasets organize sleep-related logs by power state, enabling structured analysis and easier identification of sleep and wake event trends.

9. Workflow Actions

Workflow Actions are custom actions you can add to your search results to link data to other searches, dashboards, or even external tools/websites.

By using two methods:

- **Search Method:** I configured a Workflow Action using the **Search method**, which allowed me to right-click a specific field value in the results and instantly run a new search with that value as a filter.

Destination app: search

Name: Search Sleep Events
Enter a unique name without spaces or special characters. This is used for identifying your workflow action later on within Splunk Settings.

Label: Search Sleep Events for: \$host\$
Enter the label that appears for this action. Optionally, incorporate a field's value by enclosing the field name in dollar signs, e.g. 'Search for ticket number : \$ticketnum\$'.

Apply only to the following fields: host
Specify a comma-separated list of fields that must be present in an event for the workflow action to apply to it. When fields are specified, the workflow action only appears in the field menus for those fields; otherwise it appears in all field menus.

Specify a comma-separated list of event types that an event must be associated with for the workflow action to apply to it.

Show action in: Fields menus

Action type: search

Search configuration

Search string: index="main" host="calvisitor-10-105-162-124" | stats count BY power_state
Enter the search for this action. Optionally, specify fields as \$fieldname\$, e.g. sourcetype=redis controller=\$controller\$ error=.

Run in app: search
Choose an app for the search to run in. Defaults to the current app.

Open in view:
Enter the name of a view for the search to open in. Defaults to the current view.

Run search in: New window

Time range

Earliest time: -5mon:0mon

Latest time: 5mon

Use the same time range as the search that created the field listing

Cancel Save

Results

7/8/25 8:52:05.000 AM

Event Actions

Type	Field	Value	Actions
Selected	host	calvisitor-10-105-162-124 (sleep)	
	source	Mac.log	
	sourcetype	syslog	
Event	messageType	0x0000340	
	pid	0	
	process	kernel	
	sleep_flag	1	
	tag	sleep	
Time	_time	2025-07-08T08:52:05.000+00:00	
Default	index	main	
	linecount	1	
	punct		
	splunk_server	dl-03629ccac443e1747ord-p-bmb7apunkcloud.com	

Edit Tags
Search Sleep Events for: calvisitor-10-105-162-124

index="main" host="calvisitor-10-105-162-124" | stats count BY power_state

Time range: Previous month

1,670 events (7/1/25 12:00:00.000 AM to 8/1/25 12:00:00.000 AM) No Event Sampling

Events (1,670) Patterns Statistics (0) Visualization

Timeline format Zoom Out Zoom In X Deselect

Format Show: 20 Per Page View: List

Time	Event
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 kernel[0]: AppleCmIn:systemWakeCall - messageType = 0x0000340
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 sleep source = Mac.log sourcetype = syslog
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 kernel[0]: ARPT: 790786.368522: IOPMPowerSource Information: onSleep, SleepType: Normal Sleep, 'ExternalConnected': No, 'TimeRemaining': 15462,
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 sleep source = Mac.log sourcetype = syslog
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 kernel[0]: AirPort_Brcm43xx:powerCharge: System Sleep
7/8/25 8:52:05.000 AM	Jul 8 08:52:05 calvisitor-10-105-162-124 kernel[0]: PM response took 1999 ms (54, powerd)
7/8/25 8:52:03.000 AM	Jul 8 08:52:03 calvisitor-10-105-162-124 mDNSResponder[91]: mDNS_DeregisterInterface: Frequent transitions for interface awd10 (FE80:0000:0000:0000:0000:0000:0000:0000)

- **Link Method:** I configured a Workflow Action using the Link method, enabling me to click a field value and directly open website related to that data.

Label *

Enter the label that appears for this action. Optionally, incorporate a field's value by enclosing the field name in dollar signs, e.g. Search for ticket number : \$ticketnum\$.

Apply only to the following fields

Specify a comma-separated list of fields that must be present in an event for the workflow action to apply to. When fields are specified, the workflow action only appears in the field menus for those fields; otherwise it appears in all field menus.

Apply only to the following event types

Specify a comma-separated list of event types that an event must be associated with for the workflow action to apply to it.

Show action in

Action type *

Link configuration

URI *

Enter the location to link to. Optionally, specify fields by enclosing the field name in dollar signs, e.g. [https://www.google.com/search?q=\\$host\\$](https://www.google.com/search?q=$host$).

Open link in

Link method

The screenshot shows the Windows Security Event Viewer interface. The top pane displays the event log, and the bottom pane shows the details of a selected event. The event is a 'Sleep' event (ID 4096) from the 'System' log, occurring on 2025-07-08 at 08:52:05.000+00:00. The 'Process' field is highlighted with a red box, and a tooltip 'Check process on VirusTotal: kernel' is visible.

Type	Field	Value	Actions
Selected	host	localhost (sleep)	
	source	MacLog	
	sourcetype	syslog	
Event	eventtype	sleep	
	pid	0	
	process	kernel	
	sleep_flag	1	
	tag	sleep	
Time	_time	2025-07-08T08:52:05.000+00:00	
Default	index	main	
	linecount	1	
	punct		

kernel

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COMMENTS 16k

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77bd672f2f14cab9697662443bba1cae49855fc70e13af64db3982fb76237

Posted 3 days ago

YARA Signature Match - THORAPT Scanner-br /
/RULE: SUSP_PSI_Kernel_import_Ignore_Error_May21_3-br />RULE.SET: Livehunt - Suspicious Indicators
/
/RULE_TYPE: THORAPT Scanner's rule set only
/
/RULE.LINK: https://valhalla.netron-systems.com/info/rule/SUSP_PSI_Kernel_import_Ignore_Error_May21_3-br />DESCRIPTION: Detects suspicious kernel ntldr.dll imports in PowerShell code-br />REFERENCE: https://github.com/BlackOverX/cqtools-br />RULE_AUTHOR: Florian Roth-br /> Hey, this seems to be an open-source tool or framework. The author has shared it with the community with the intention to improve overall security. If you are a victim and noticed this tool in a breach, please visit the tool's github page (see above) and tell your story by creating an issue on the issues page: https://github.com/BlackOverX/cqtools/issues
/
/Show more

579dcab5bca14684afdee1959c68ff04c18b9be5c32e3ef967da3939546303c

Posted 3 days ago

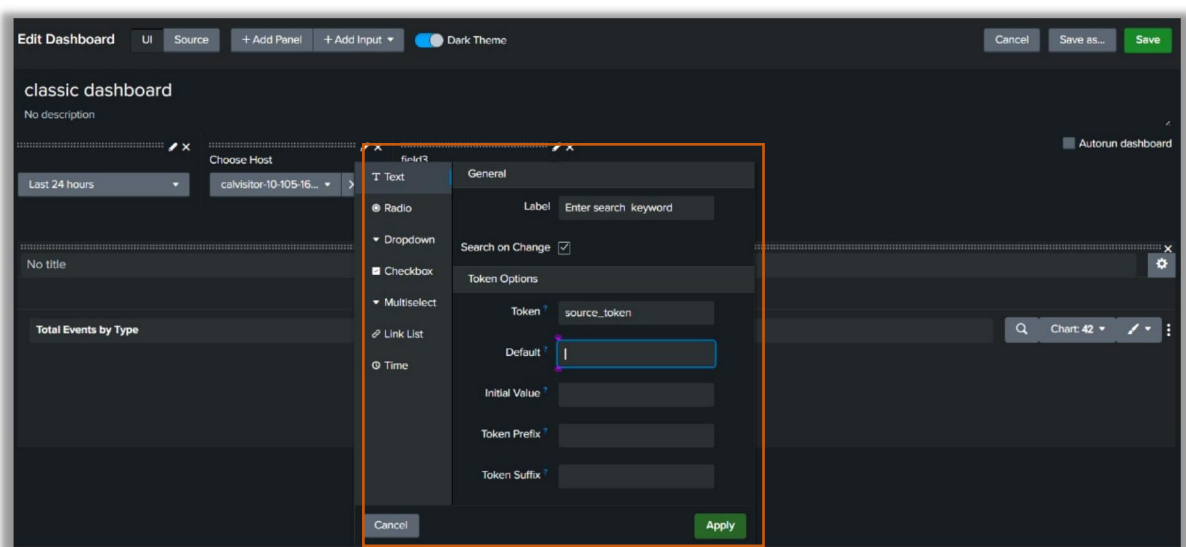
YARA Signature Match - THORAPT Scanner-br /
/RULE: HTKL_Error_Kernel_BaseAddress-br />RULE.SET: Livehunt - Hacktools Indicators
/
/RULE_TYPE: VALHALLA rule feed only
/
/RULE.LINK: https://valhalla.netron-systems.com/info/rule/HTKL_Error_Kernel_BaseAddress-br />DESCRIPTION: Detects a hacktool that reports error while retrieving Kernel base address-br />REFERENCE: https://github.com/Darhton/Blackbone-br />RULE.AUTHOR: Florian Roth-br /> Hey, this seems to be an open-source tool or framework. The author has shared it with the community with the intention to improve overall security. If you are a victim and noticed this tool in a breach, please visit the tool's github page (see above) and tell your story by creating an issue on the issues page: https://github.com/Darhton/Blackbone/issues
/
/Detection.Timestamp: 2025-08-06 04:15:00 />AV Detection Ratio:
/
/Show more

The workflow action provides direct navigation to Virustotal for efficient investigation of processes.

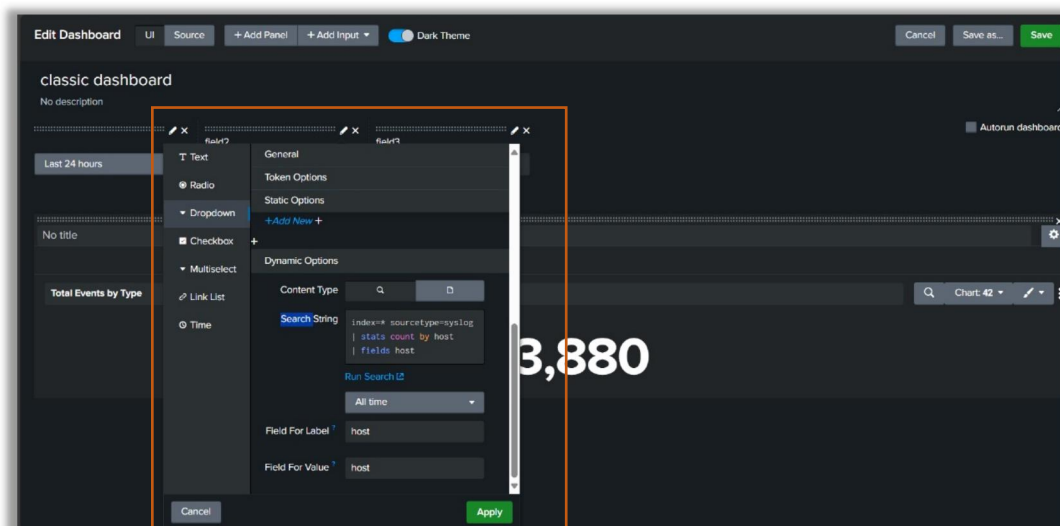
10. Input Controls in Dashboard

Input controls such as dropdowns and text inputs make dashboards interactive by providing filters that users can select or enter.

- **Text Input Control:** I set up a text input box to enable users to enter keywords or source names. The input captures user entries via a token, allowing flexible and dynamic search queries that filter log data according to the entered text, enhancing dashboard usability.



- **Dropdown Input Control:** I configured a dropdown input in the Splunk dashboard to allow dynamic selection of hosts. This input uses a token to capture the selected host value, enabling interactive filtering of data based on host, which can be linked to dashboard panels for real-time analysis.

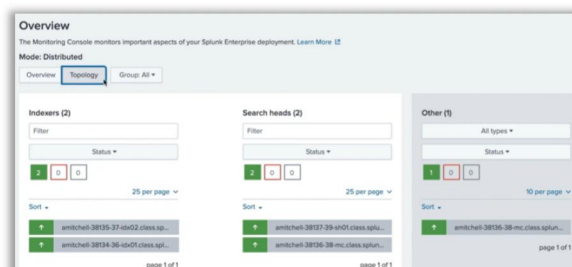
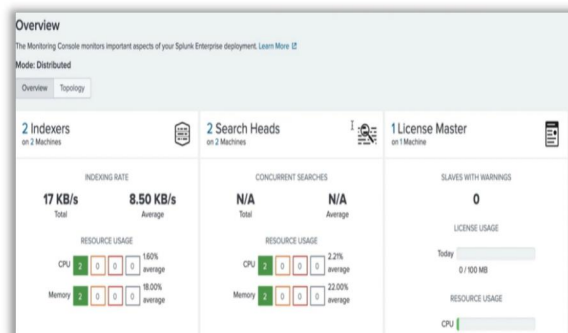


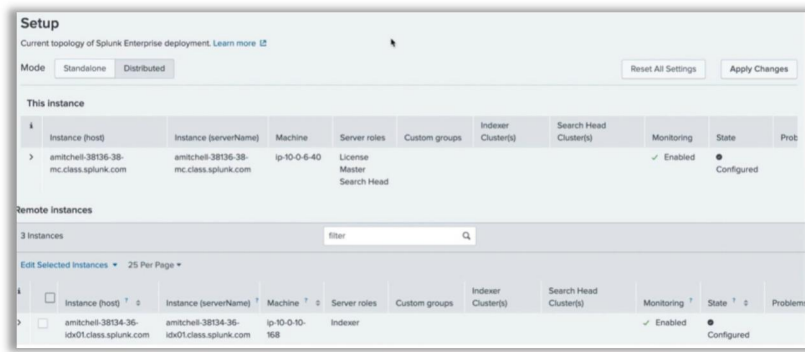
Search Job properties	
canSummarize	None
createTime	2025-08-07T17:05:31.000-08:00
cursorTime	1970-01-01T00:00:00.000-08:00
custom	([])
defaultEventTTL	604800
defaultTTL	600
delegate	None
diskUsage	180224
dispatchState	DONE
doneProgress	1
dropCount	None
esIndex	([])
earliestTime	2025-08-07T00:00:00.000-08:00
eventAvailableCount	1000

Findings: Job Inspector provides complete details of each search query including job properties and execution components, enabling in-depth analysis of search performance and efficient troubleshooting of query issues.

12. Monitoring console

A centralized dashboard in Splunk used to track system health, resource usage, indexing performance, and overall deployment status.





Findings: The Monitoring Console's dashboards for indexer, search head, license manager, instances, and setup provide centralized visibility into system performance, license usage, and cluster health, enabling effective monitoring and timely issue resolution.

Note: Monitoring Console screenshots are sourced from the official Splunk website for reference and better understanding.